Project Synopsis - Submission

HCLTech Career Shaper Programme | III Year Industry Project | 2025-26

Field	Details
Project Title	Supply Chain Analytics and Transportation Optimization System (OR-Tools / PuLP)
Project No.	[FILL IN - template shows 48]
Domain	Data Analytics / Python / Operations Research
Duration / Format	[FILL IN] / Individual Project
Learner Name	[FILL IN]
Roll Number	[FILL IN]
College / Campus	[FILL IN]
Mentor Name	[FILL IN]
Submission Date	[FILL IN]
GitHub Repository	[PASTE YOUR GITHUB REPO LINK HERE]

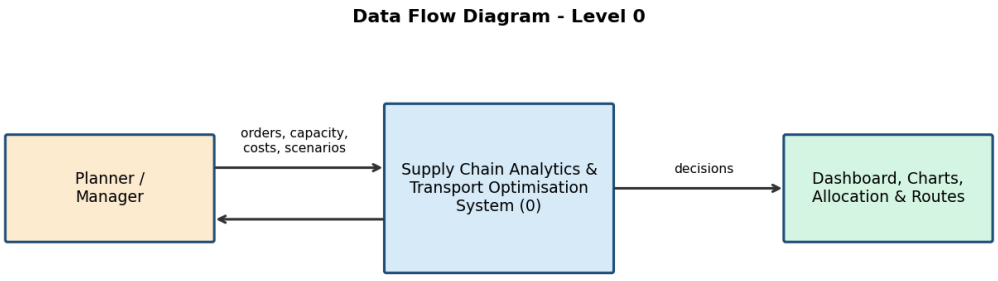
1. Problem Statement

Distribution managers at multi-warehouse companies must decide which warehouse should serve which customer and how trucks should be routed, usually with limited warehouse stock and rising transport costs. When these decisions are made by habit or by the nearest-warehouse rule, transport cost and distance driven are higher than necessary, and peak-season demand (Oct-Dec in this project) strains capacity. This project analyses shipment history and uses mathematical optimisation to reduce transport cost and kilometres driven while still meeting all demand.

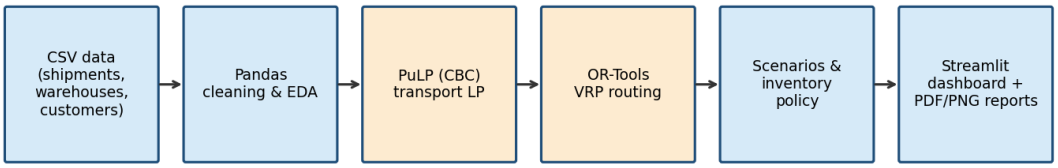
2. Proposed Solution & Approach

I built a Python pipeline with four parts. (1) Pandas analytics on 24 months of shipments (demand trend, cost, delays). (2) A PuLP linear programme that allocates stock from 6 warehouses to 18 customers at minimum cost under capacity limits. (3) An OR-Tools capacitated vehicle-routing model that builds truck routes from each warehouse. (4) A safety-stock and reorder-point table for each customer. The optimised plan is compared with a nearest-warehouse baseline and with direct trips.

3. System Architecture



System Block Diagram



4. Technology Stack

Layer	Used
Frontend / UI	None (command-line pipeline; PDF and PNG reports)
Backend / Logic	Python 3.12, Pandas, NumPy
Database	CSV files in data/
Optimisation	PuLP 2.9.0 (CBC solver), Google OR-Tools (routing)
AI/ML Framework	Not applicable (operations research, not machine learning)
Visualisation	Matplotlib
Hardware	Standard laptop; no special hardware

5. Dataset Used

Important: no real dataset was available, so the data is **synthetic**, generated by *src/generate_data.py* (random seed 42). City coordinates are approximate; demand, costs and delays are simulated. Results demonstrate the method, not real company savings.

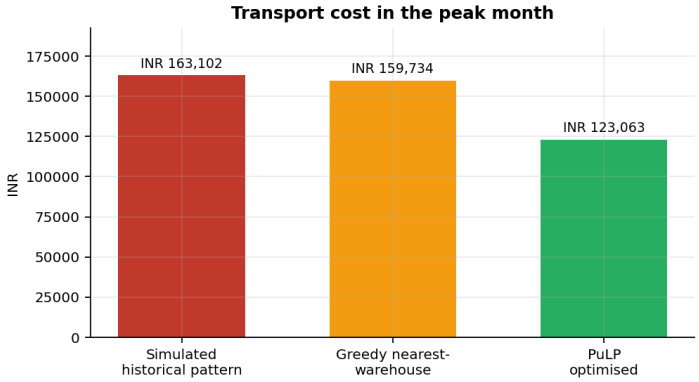
Item	Value
Main file	data/shipments.csv (plus warehouses.csv, customers.csv, distance_matrix.csv)
Rows (shipments.csv)	4,363
Columns (shipments.csv)	13
Key columns	order_date, warehouse, customer, product_category, priority, quantity, distance_km, transport_cost, delayed
Additional data collected	None

6. Key Results & Metrics

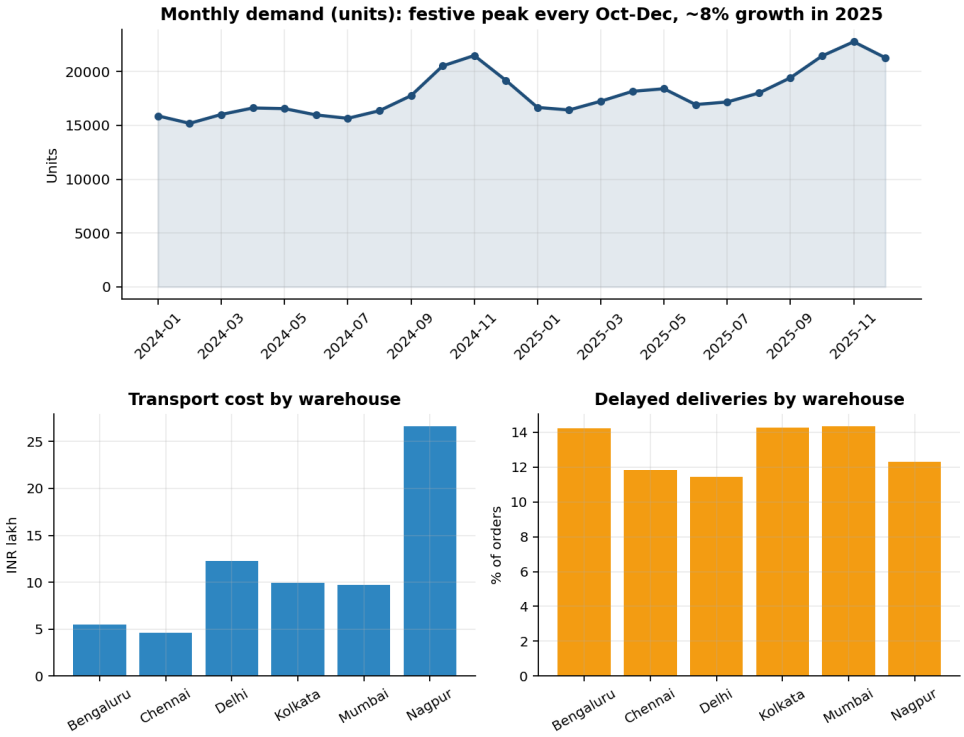
Metric	Value
Peak-month demand planned (month 11, 2025)	22,778 units
Transport cost: greedy nearest-warehouse baseline	INR 159,734
Transport cost: PuLP optimised	INR 123,063 (22.96% lower than baseline)
Transport cost: simulated historical pattern	INR 163,102 (optimised is 24.55% lower)
Distance driven: direct trips vs OR-Tools routes	44,379 km vs 41,566 km (6.34% less)
Delayed deliveries in history	12.9% of orders

Business value: a cost-aware allocation and route plan lowers transport spend and kilometres driven while meeting all demand. At the assumed truck cost of INR 28 per km, the shorter routes would save about INR 78,758 in the peak month (assumed rate, simulated data). The saving depends on how tight warehouse capacity is (about 84% utilised at peak) and on how weak the baseline is; the simulated history also ships about 40% of orders from farther warehouses by design, so the

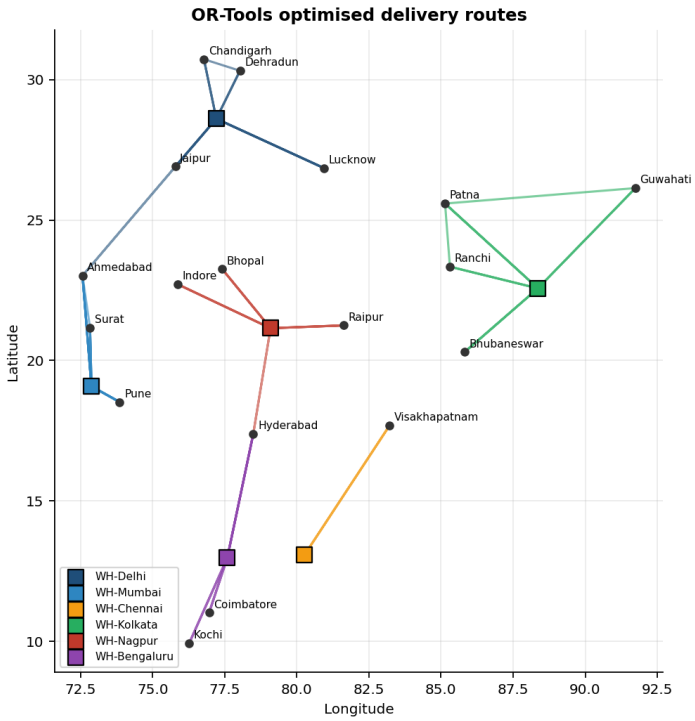
greedy comparison is the more conservative one.

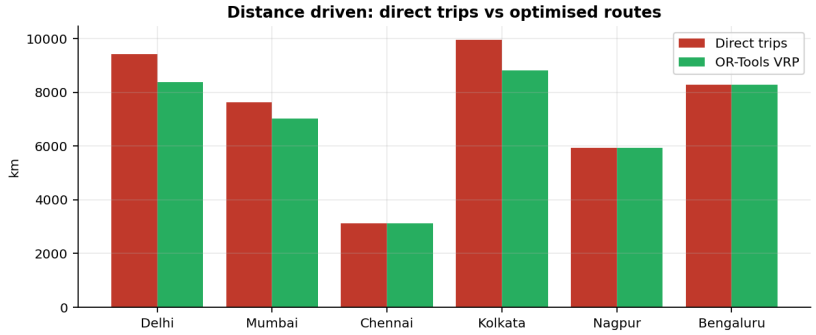


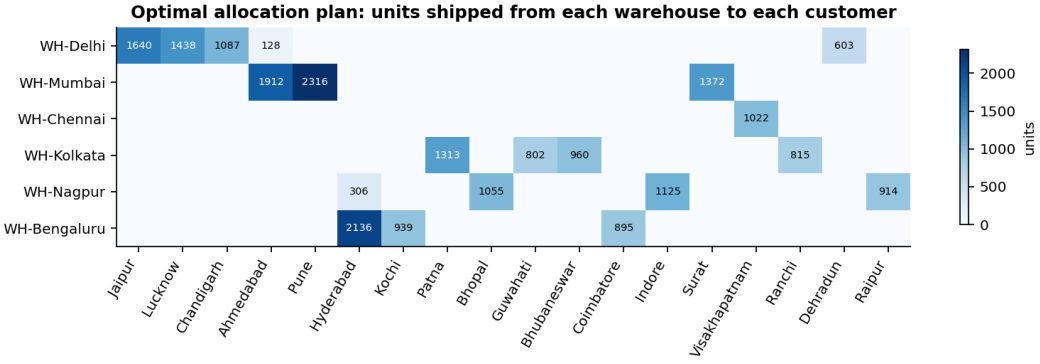
Analysis & Optimisation Outputs

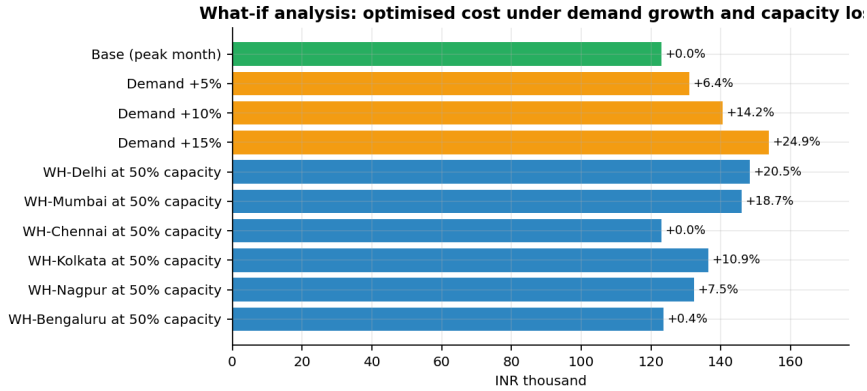


Demand peaks in October-December. Delay rates differ by warehouse and distance; because delays are simulated, I do not draw real-world conclusions from that pattern.









Routing summary

Warehouse	Customers	Trucks	Direct km	Optimised km
WH-Delhi	5	9	9418	8381
WH-Mumbai	3	10	7635	7024
WH-Chennai	1	2	3132	3128
WH-Kolkata	4	7	9960	8819
WH-Nagpur	4	7	5944	5934
WH-Bengaluru	3	8	8291	8280

Optimal allocation (8 largest flows, units)

Warehouse	Customer	Units
WH-Mumbai	Pune	2316
WH-Bengaluru	Hyderabad	2136
WH-Mumbai	Ahmedabad	1912
WH-Delhi	Jaipur	1640
WH-Delhi	Lucknow	1438

WH-Mumbai	Surat	1372
WH-Kolkata	Patna	1313
WH-Nagpur	Indore	1125

Inventory policy (first 6 customers; full table in outputs/inventory_policy.csv)

Customer	Main Warehouse	Lead Time Days	Avg Daily Demand	Safety Stock	Reorder Point
Jaipur	WH-Delhi	1.7	39.6	13	79
Lucknow	WH-Delhi	2.2	35.8	12	91
Chandigarh	WH-Delhi	1.7	28.3	9	57
Ahmedabad	WH-Mumbai	2.3	54.0	18	141
Pune	WH-Mumbai	1.3	56.7	17	93
Hyderabad	WH-Bengaluru	2.4	67.8	23	188

7. Challenges Faced & How Resolved

Challenge 1: no real dataset was available. **Resolution:** generated a documented synthetic dataset (fixed seed, realistic seasonality) and disclosed this clearly.
Challenge 2: some customers' demand exceeded one truck's capacity, which a basic routing model cannot serve. **Resolution:** split such demand into full-truckload visits before solving the routing model.
Challenge 3: the newest PuLP release changed its API and the code failed. **Resolution:** pinned PuLP 2.9.0 in requirements.txt.

8. References

1. PuLP documentation, COIN-OR project (coin-or.github.io/pulp).
2. Google OR-Tools documentation, Routing section (developers.google.com/optimization).
3. Hillier, F. S. and Lieberman, G. J., *Introduction to Operations Research*, McGraw-Hill (the transportation problem).
4. Toth, P. and Vigo, D. (eds.), *Vehicle Routing: Problems, Methods, and Applications*, SIAM.
5. Chopra, S. and Meindl, P., *Supply Chain Management: Strategy, Planning, and Operation*, Pearson (safety stock).
Please check edition and year against your library before submitting.

9. Ethical Compliance Declaration

I confirm that:
[ ] All dataset usage complies with the source licence (data is self-generated and synthetic).
[ ] No personally identifiable information (PII) is stored.
[ ] AI tools used for code/writing are disclosed in the Appendix.
[ ] This project complies with the DPDP Act 2023 where applicable.

Learner Signature: _____ Date: _____

Appendix: AI Disclosure & How to Run

AI disclosure: the code structure, synthetic data generator and report text were drafted with the help of an AI assistant (Claude by Anthropic) and reviewed by the learner. [Edit this line to describe your own contribution accurately.]

GitHub: [PASTE YOUR GITHUB REPO LINK HERE]
Run: `pip install -r requirements.txt` then `python main.py`